Analysis Plan v1.1 — Internal Prespecified Re-analysis Plan for Resubmission

Study: ThyroidXL preoperative malignancy prediction and SEER nodal burden and age modelling in thyroid carcinoma (title to be set after decision D1)
Version: v1.1, issued 24 Oct 2026, after the ThyroidXL and SEER audits and **before any outcome model is re-run**.
Supersedes: v1 (same date). v1 entries are kept in the deviation log (Section 9).
Context: The previous submission was rejected. Reviewer comments are used as an internal quality-control checklist (Section 10), not as a revision to answer. Manuscript text changes come last.
Audit record: "ThyroidXL Audit — Findings and Decisions" (Sections 1–6), which documents every count used below.

Transparency note: Results of the previous analysis (v1.0.1) are known to the authors. All decisions below were taken after the data audits and before re-running any outcome model. Previous LNR/LODDS results are superseded (Section 3.13).

1. Scientific message

Level	Message
Primary	In ThyroidXL, ultrasound-based ML models give a modest, internally cross-validated gain in discrimination over ACR-TIRADS; robustness is tested in the histopathology-confirmed subset.
Secondary	In SEER differentiated thyroid carcinoma (DTC), LNR and LODDS are compared with AJCC-8 N category for short-term cancer-specific survival under an identical adjustment set, with optimism-corrected performance.
Integrative context	Age-gradient comparison across the two cohorts, presented as contextual synthesis, not as a main result. Retained after the prespecified feasibility check (Section 4; D1 = one paper). Framing: an integrated two-stage risk-stratification study, not a "bridge study".

2. Diagnostic analysis — Cohort A (ThyroidXL)

2.1 Population. N = 4,088 (4,093 minus 5 with unrecoverable age). One nodule per patient; one row per patient.

2.2 Outcome (primary): reference-standard hierarchy.

- Histopathology overrides FNAC wherever a histopathology text is available; otherwise the released `conclusion` code is used (1, 2 = malignant; 3, 4 = benign).
- Changes 4 labels relative to the released dataset: patients 256, 509 and 370 become malignant; patient 366 becomes benign. Primary malignant count = 1,231.
- Follicular thyroid tumor (2 patients, 1548 and 3919): kept non-malignant; histology is not definitive for malignancy.
- Histopathology-coded cases without text (9): keep their code label.
- FNAC-benign labels with Bethesda III or V cytology (9): kept; reported descriptively.
- `conclusion = 4` is histopathology-confirmed benign. The terms "suspicious", "indeterminate" or "unresolved" are not used for these cases.
- **Sensitivity:** labels as released (1,229 malignant).

2.3 Predictors (Set A). tirads, age (continuous), gender, width, height, nodule_area, aspect_ratio, pos_isthmus, pos_left. `age_55_plus` is removed as redundant with continuous age.
Sensitivity: v1.0.1 feature set (with `age_55_plus`), to show that its removal does not drive results.

2.4 Models. Logistic regression (C = 1.0), Random Forest (100 trees, max depth 6, min leaf 20, balanced weights), XGBoost 2.1.3 (100 trees, learning rate 0.05, subsample 0.8), compared with ACR-TIRADS. Hyperparameters are fixed and prespecified; no tuning.

2.5 Validation pipeline.

- Stratified 5-fold CV with one fixed fold assignment (fixed seed) shared by all models and TIRADS.
- Scaling and TIRADS empirical malignancy rates fitted on the training fold only; no imputation (complete cases defined before CV).
- Output: one out-of-fold (OOF) prediction per patient per model.
- The v1.0.1 code was reviewed and no leakage was found (audit Section 2.8). A pipeline schematic is added for reporting.
- Grouped folds are not needed (one nodule per patient).

2.6 Metrics.

- AUC with 95% CI (2,000 bootstrap resamples of OOF predictions).
- Paired ΔAUC vs TIRADS: identical resamples for both members of a pair; 95% percentile CI; DeLong paired test as confirmatory.
- Calibration: intercept, slope, smoothed calibration curve, Brier score.
- Secondary: Sens/Spec/PPV/NPV with Wilson CIs at TIRADS ≥4 and at the ML threshold matching TIRADS ≥4 sensitivity.

2.7 Reference-standard sensitivity analyses.

- **S-D1 histopathology-confirmed subset** (main stronger-reference analysis): 796 patients, 704 malignant, 92 benign; the 2 follicular thyroid tumors are excluded.

    - Outcome derived solely from histopathology text for every case, independent of the original coding.
    - Role: supportive sensitivity analysis for all Set A models; not primary evidence.
    - Reported: AUC with 95% CI, paired ΔAUC vs TIRADS with 95% CI, calibration descriptively.
    - No claim that performance is "retained" unless the CIs and paired differences support it; no direct comparison with full-cohort AUC as deterioration or improvement (surgical selection; 79/92 benign are thyroid cyst disease).
    - Discordance between released labels and histopathology-derived labels is reported (audit Section 2.4).
- **S-D2** (histopathology, else Bethesda II/VI): 4,072 patients. Reported in one line as a supporting check only, stating that it nearly equals the full cohort because FNAC labels are almost all Bethesda II or VI.
- **Former S-D3 / Table S15 is removed**: it excluded histopathology-confirmed benign cases and answers no valid question.

2.8 Set B (FNAC as predictor). Supplementary only, explicitly labelled as subject to incorporation bias; full-cohort AUC is not presented as evidence of independent value.

- Within S-D1, Set B is reported inferentially only if ≥100 malignant and ≥100 benign cases are present. This criterion was prespecified by analogy to published guidance on sample size for reliable performance estimation in model validation (Vergouwe et al., J Clin Epidemiol 2005; Collins et al., Stat Med 2016); it is not presented as a rule specific to internal CV.
- **Result of the audit:** S-D1 with FNAC has 566 malignant and 88 benign, so Set B in S-D1 is **descriptive only**.
- Manuscript wording: "Set B was evaluated inferentially only when both outcome classes were sufficiently represented for stable performance estimation; otherwise it was reported descriptively."

2.9 Bethesda categories.

- Descriptive table: Bethesda × label source × final label (audit Section 2.3).
- Bethesda III (n = 39, 26 malignant): exploratory, no between-model claims.
- Bethesda IV (n = 6) and V (3 benign): not analysed for discrimination. Pooled III–V is not a headline result.
- Stated explicitly: ThyroidXL is not suitable for establishing model performance in a representative indeterminate-cytology population.

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3. Prognostic analysis — Cohort B (SEER 2018–2022)

3.1 Data corrections (audit Section 6), applied before any cohort definition.

- **S1 Nodal special codes:** examined 95/96/97 and positive 95/97 are non-quantitative and set to missing for LNR/LODDS/N modelling. These records leave the quantitative nodal cohort and are reported as a separate descriptive group (all histologies: 840 records, 145 CSS events; DTC: 675 records, 56 events; counted in the full cleaned file before survival-time and cause-of-death filtering. The audit's earlier figures, 817 and 656, used a narrower rule that also required the other nodal field to be <98, and are superseded). Reported in Methods, flow diagram and Limitations.
- **S2 Examined = 90** ("90 or more"): kept as 90, described as a lower-bound approximation. Sensitivity: exclusion of these records.
- **S3 Histology map (ICD-O-3):**
    - PTC: 8050, 8260, 8337, 8339, 8340, 8341, 8342, 8343, 8344, 8350
    - FTC (including oncocytic): 8290, 8330, 8331, 8332, 8335
    - MTC: 8345, 8510; mixed medullary: 8346, 8347 (not DTC)
    - ATC: 8020, 8021, 8022, 8030–8035
    - Other: all remaining codes (including 8000/8001/8010 NOS)
    - All subtype counts are re-derived from the raw file; no count from the previous manuscript is reused.
- **S4 Unknown cause of death (118):** handled per Section 3.3.

3.2 Population.

- **Primary:** DTC (PTC + FTC as defined in 3.1) with quantitative nodal data (examined 1–90, positive 0–90, positive ≤ examined) and available survival time. Expected before covariate filtering: ≈31,490 patients, ≈258 CSS events.
- Flow diagram from 61,615 with exact counts at each step: special nodal codes, no nodes examined, missing survival time (141, all deceased), unknown cause of death, missing covariates.

3.3 Endpoint. Cancer-specific survival, framed as short-term. Horizons: 12 and 24 months primary; 36 months exploratory.

- **Primary:** records with unknown cause of death excluded from cause-specific analyses; number reported.
- **Sensitivity A:** unknown-COD deaths counted as CSS events.
- **Sensitivity B:** unknown-COD deaths treated as non-thyroid deaths (censored at death in cause-specific Cox; competing events in Aalen–Johansen).

3.4 Covariates and coding (S5, fixed before fitting).

- Age continuous (linear); sex; histology (PTC/FTC).
- T: T1a + T1b → T1; T2; T3a + T3b → T3; T4a + T4b → T4 (3 df).
- M: M0 / M1 (1 df).
- N category: N0 / N1a / N1b, with N1 NOS as its own category if present (≤3 df).
- TX, MX, NX, unknown and missing are not merged into any category; they follow the missing-data rule: complete case primary, multiple imputation sensitivity if >5% of the cohort is excluded for missing covariates.
- Not included: overall AJCC stage, `age_55_plus`, tumour size.
- Treatment variables (surgery type, RAI): sensitivity analysis only.

3.5 Models.

- **M0** Core (age, sex, histology, T, M).
- **M1** Core + N category.
- **M2** Core + LNR (continuous).
- **M3** Core + LODDS (continuous).

3.6 Estimation.

- Events per parameter are computed **after all exclusions and complete-case restriction**, for the largest model (M1).
- If EPP ≥20: unpenalised Cox is primary for all four models.
- If EPP <20: ridge Cox is primary for all four models, with λ selected by internal CV and re-selected in every bootstrap resample.
- The other approach is reported as sensitivity. The fixed λ = 0.1 is retired.
- Age non-linearity: RCS (3 knots, 10/50/90th percentiles) sensitivity.

3.7 Discrimination. Harrell's C and Uno's C truncated at τ = 24 months (primary: Uno's C at τ = 24); time-dependent AUC at 12 and 24 months (36 exploratory). Optimism correction by Harrell bootstrap (B = 500, full refit including any λ selection). Paired ΔC with identical resamples and 95% percentile CI.

3.8 Calibration (diagnose before recalibrating).

1. 24-month calibration slope, O:E ratio, decile plot, optimism-corrected.
2. Diagnostic comparison: penalised vs unpenalised; all-histology vs DTC.
3. Recalibration only if the validated slope remains outside 0.8–1.2 after step 2; reported as recalibrated with bootstrap-validated slope.
4. If calibration remains poor, models are described as risk-ranking only.

3.9 Proportional hazards. Schoenfeld tests and plots; if violated, HRs reported as averaged effects and inference rests on time-dependent AUC or landmark analyses.

3.10 Secondary and descriptive.

- All-histology models (to show the influence of ATC).
- ATC, MTC, FTC descriptive (n, events, KM); no model comparison where events are few.
- The special-nodal-code group described separately (n, events, histology mix).
- D ≥5 restriction, LNR cut-points, competing-risk CIF: Supplementary.

3.11 Decision curve analysis. Only if the validated calibration slope is within 0.8–1.2; otherwise removed.

3.12 Sensitivity analyses (summary). Unknown COD A and B; examined = 90 excluded; treatment variables added; RCS age; ridge or unpenalised (whichever is not primary); multiple imputation if triggered.

3.13 Previous results. All previous LNR/LODDS/N-stage results (C-indices, HRs, calibration, DCA) are superseded and are not reused or cited as findings. ThyroidXL diagnostic results are unaffected by the SEER corrections.

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4. Bridge feasibility check (prespecified; decides D1)

Question: Do independent diagnostic and prognostic cohorts show reproducible concordant or discordant population-level risk gradients across shared strata?

4.1 Populations.

- SEER: DTC primary; all-histology as reference; PTC-only sensitivity.
- ThyroidXL: full cohort (primary outcome rule); S-D1 sensitivity.
- PTC-only is not defined on the ThyroidXL side (521 FNAC-malignant cases have no subtype).

4.2 Strata. Start from <35, 35–44, 45–54, 55–64, 65–74, ≥75. A stratum is kept only if it has ≥10 CSS events in SEER DTC and ≥30 ThyroidXL patients; otherwise it is merged with the adjacent stratum, merging toward the middle, until both conditions hold. The same final strata are used in both cohorts.

4.3 Endpoints.

- ThyroidXL: malignancy proportion with Wilson 95% CI.
- SEER: 24-month CSS cumulative incidence by Aalen–Johansen (other-cause death as competing event), with 95% CI. Crude proportions are not used.

4.4 Estimands.

- Within-cohort age slopes only: ThyroidXL logistic OR per 10 years (sex-adjusted); SEER DTC Cox HR per 10 years (sex-adjusted).
- No correlation or test across cohorts, no rank correlation across strata.
- Age × sex: 2 × 2 (<55 / ≥55 × sex), descriptive with CIs.

4.5 Decision rule. One paper is kept only if all three hold:

- (a) **Estimability:** all final strata meet the 4.2 rule.
- (b) **Stability** (exclusion of the null is not required):
  - (b1) Direction: the sign of the age effect is the same in the primary and sensitivity analyses of each cohort: ThyroidXL full cohort and S-D1; SEER DTC and PTC-only.

    - (b2) No large reversal: for each primary estimate (ThyroidXL full cohort; SEER DTC), the 95% CI limit on the opposite side of 1 from the point estimate lies within 0.91–1.10 per 10 years, i.e. the CI does not admit a reverse-direction effect of ≥10% per decade.
    - A CI that includes 1 does not fail (b) if (b1) and (b2) hold. A sign change between analyses, or a CI admitting a ≥10% per decade reverse effect, fails (b).
- (c) **Framing:** the integrated message can be stated as complementary within-cohort findings without cross-cohort inference.

Otherwise the work is split. In either case, the feasibility results are reported (in the single paper or in the SEER paper).

4.7 Result (24 Oct 2026). Final strata <45, 45–54, 55–64, 65–74, ≥75. (a) met. (b) met: ThyroidXL OR per 10 y 0.77 (0.73–0.81), S-D1 0.86 (0.72–1.02); SEER DTC HR per 10 y 2.44 (2.28–2.62), PTC-only 2.49 (2.31–2.69). (c) met by author judgement with the following framing:

> Older age was associated with lower odds of malignancy in the diagnostic cohort and higher short-term cancer-specific mortality in the prognostic cohort. These complementary within-cohort associations do not establish an individual-level diagnostic-to-prognostic transition and require confirmation in patient-linked cohorts.

The attenuated S-D1 estimate is reported alongside the full-cohort estimate. **D1 = one paper.** The bridge is not used as the justification for keeping one paper.

4.6 Interpretation limits. Ecological and descriptive only; crude malignancy proportions are not mixed with time-specific mortality; referral pattern, geography, ascertainment, sex distribution, histology mix and prevalence are listed as alternative explanations; no individual-level or causal interpretation.

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5. Comparison and interpretation rules

1. Any statement that X performs better than Y requires a paired difference with 95% CI. CI excluding 0 → "higher"; otherwise "numerically higher" or "comparable".
2. **Primary families (Holm-adjusted):**
    - Diagnostic: LR, RF, XGBoost each vs TIRADS (ΔAUC, full cohort). All three are co-primary; no single model will be promoted as primary based on observed performance.
    - Prognostic: M2 vs M1 and M3 vs M1 (Δ Uno's C at τ = 24, optimism-corrected).
3. All other comparisons: nominal CIs, labelled secondary or exploratory.
4. Apparent and validated estimates shown side by side; the Abstract uses validated estimates only.
5. Bootstraps: resampling unit = patient; fixed seeds; identical resamples across compared models.
6. Statistical significance is not described as clinical importance.
7. Any analysis added after this plan is labelled post hoc and logged in Section 9.

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6. Prohibited claims

- ML is clinically superior to, or should replace, ACR-TIRADS.
- Clinical utility from DCA unless 3.11 conditions are met.
- Absolute-risk prediction if calibration remains poor.
- Long-term prognosis, late recurrence or long-term survival.
- Value of the models in indeterminate (Bethesda III–IV) nodules or post-cytology risk refinement.
- Diagnostic-to-prognostic continuum, biological transition, pathway or causal language for the cross-cohort component.
- Any individual-level statement linking the cohorts (e.g. that a malignant nodule in an older patient is more lethal); the cohorts are independent.
- Generalisability beyond internal validation.
- Any description of `conclusion = 4` cases as suspicious, indeterminate or unresolved.

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7. Reproducibility

- New release v1.1.0 after all analyses are complete: corrected outcome hierarchy; corrected SEER cleaning (S1–S4); audit scripts and outputs; paired ΔAUC; S-D1; DTC survival analyses with optimism correction and paired ΔC; calibration and time-dependent analyses; bridge feasibility check; updated verification script.
- Zenodo record created only after all results and files are fixed.
- v1.0.1 stays archived as the record of the previous analysis.

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8. Decision register

#	Decision	Status
D1	One paper or two	**One paper** (Section 4.7); integrated two-stage risk-stratification framing
D2	Target journal	After D1 and new results; shortlist started; not the previous framing at the previous journal
D3	Plan framing	Internal re-analysis plan for resubmission

D4	Primary outcome rule	Histopathology overrides FNAC; released labels as sensitivity
D5	Former S-D3 / Table S15	Removed
D6	S-D2	One-line supporting check; S-D1 is the main stronger-reference analysis
D7	Follicular thyroid tumors	Non-malignant in primary; excluded from S-D1
D8	Set B	Supplementary; descriptive in S-D1
D9	Bethesda	No indeterminate-nodule claims
D10	Bridge	Reframed question; conditional on Section 4
D11	Reproducibility	v1.1.0 release after analyses
D12	`age_55_plus`	Removed from Set A; old feature set as sensitivity
A1	FNAC-benign labels with Bethesda III/V (9)	Kept; descriptive
A2	Histopathology-coded cases without text (9)	Keep code label
A3	992 extra patients in `new_id2info.json`	Not used (not in the released image dataset)
A4	Age birth-year reference 2022	Check the ThyroidXL paper; keep 2022 if no year is reported, and state it
S1–S5	SEER corrections	Adopted (Section 3.1, 3.3, 3.4)

9. Deviation log

Date	Section	Change	Reason	Before/after results seen
24 Oct 2026	v1: 2.7, 2.8, 3.1, 3.4, 5	S-D2 deferred to audit; Set B threshold justified; DTC defined; EPP condition and fallback; co-primary statement; Set B criterion framed as by-analogy, reporting split	Internal review of v1	Before (no new analysis run)
24 Oct 2026	v1: 2.7	S-D1 role and reporting rules; S-D2 non-construction rule; S-D1 outcome re-derived from histopathology with discordance reporting	Aggregate figures (798 / 708 / ≈90)	Before
24 Oct 2026	v1.1: whole plan	Framing changed to resubmission; outcome hierarchy (D4); Set A without `age_55_plus` (D12); S-D3 removed; SEER corrections S1–S5; DTC map redefined; unknown-COD rules; T collapsed; bridge feasibility check added; previous LNR/LODDS results superseded	ThyroidXL and SEER audits; rejection of previous submission	After audits, before any outcome model re-run
24 Oct 2026	v1.1: 4.5(b)	Stability criterion changed from null exclusion to direction consistency plus a bound on reverse-direction effects (≥10% per decade); A1–A4 confirmed	Internal review of v1.1; decision rule should not reduce to significance testing	Before any outcome model re-run
24 Oct 2026	1, 4.7, 6, 8	D1 decided (one paper) after the prespecified check; framing sentence fixed; individual-level linkage added to prohibited claims	Section 4 result; criterion (c) by author judgement	After the bridge check; before diagnostic and prognostic model runs
24 Oct 2026	3.9	Post hoc histology-stratified Cox (FTC PH violation, $p < 0.001$ in all models); sensitivity only, does not replace primary models	FTC PH violation and FTC coefficient change after adding N	Identified after stage-1 diagnostics; added before stage-2 inferential results
24 Oct 2026	2.1 / A4	Birth-year age reference year 2022 → 2024 (ThyroidXL paper: collection over two years from Feb 2023; no per-patient year); sensitivities 2023, 2025, exclusion; 34 records affected	Documentation check; 2022 predates data collection	After diagnostic results; effect on AUC ≤ 0.001, conclusions unchanged
24 Oct 2026	3.6–3.12	Implementation: Efron Cox engine validated against lifelines; survival time 0 → 0.5 month; paired corrected Δ CI = bootstrap-sample Δ percentile interval shifted by mean Δ optimism; Uno IPCW from evaluation data; sensitivities as apparent ΔC with refit bootstrap (B = 200); ridge λ by 5-fold cross-validated partial likelihood; DCA from 10-fold cross-validated risks	Specification of methods not fixed in v1.1	Before stage-2 runs
24 Oct 2026	3.1 (S1)	Special-code counts reconciled to one definition (examined 95/96/97 or positive 95/97, full cleaned file): 840 / 675 records, 145 / 56 CSS events; replaces audit figures 817 / 656	Inconsistent counts between audit, report and checklist	After stage-2 runs; counting only, no analysis affected

10. Quality-control checklist (previous reviewer comments)

Comment	Addressed by
R1-1 Clinical applicability	2.6, Section 6
R1-2 Misclassification / incorporation bias	2.2, 2.7, 2.8, audit Section 2
R1-3 Age-only adjustment	3.4, 3.5
R1-4 Histology mixing	3.1 (S3), 3.2, 3.10
R1-5 Bridge	Section 4
R2-M1 Reference standard	2.2, 2.7
R2-M2 Leakage	2.5, 2.6
R2-M3 Bethesda	2.9
R2-M4 Optimism, penalisation	3.6, 3.7
R2-M5 Calibration	3.8, 3.11
R2-M6 Follow-up	3.3
R2-M7 Bridge	Section 4
R2-M8 Focus	Section 1, D1
R2 minor 3 Nodal cohort selection	3.1 (S1), 3.2 flow diagram
R2 minor 4–6	3.10, 3.6, Section 5
R2 minor 10 TRIPOD	Checklist mapped to this plan
R2 minor 11 Reproducibility	Section 7

11. Agreed manuscript framing (24 Oct 2026, after frozen results and 106/106 verification)

- **Working title:** Two-Stage Risk Stratification in Thyroid Carcinoma Using Ultrasound and Nodal Metrics. The two-stage framing is developed in the Abstract and Introduction.
- **Diagnostic message:** ML models improved discrimination over ACR-TIRADS (ΔAUC ≈ 0.02, internally validated) but did not improve specificity at the matched guideline sensitivity, limiting evidence for reduced unnecessary biopsy at that operating point.
- **Prognostic message:** LNR and LODDS were associated with cancer-specific mortality but added no incremental short-term prognostic discrimination beyond age, sex, histology, T and M; this is the most consistent prognostic finding across analyses. They discriminated less well than AJCC N category, except in patients with ≥5 examined nodes; the N-category gain over the core model was small and not robust. Calibration was good.
- **Central Discussion theme:** association versus incremental predictive value.
- **Integrative context:** "within-cohort associations only" wording kept identical from Abstract to Conclusion.
- **Literature positioning:** findings contrast with reports suggesting added value of LNR/LODDS, particularly where adjustment and validation were less stringent; no broad claim of contradicting the literature.
- **Journal strategy:** a journal that values methodological rigour and neutral findings; chosen before writing.